PHYS 230 - Electricity and Magnetism
Lecture 24

Instructor: Saeed Rastgoo

UNIVERSITY OF ALBERTA

Outline

- Review

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman
Physics for Scientists and Engineers, Hawkes, et al.
Physics for Scientists and Engineers With Modern Physics, Serway, Jewett

Electric Potential

- Formula 1 (only for a point charge)

$$V = \frac{1}{4\pi\epsilon_0}\frac{q}{r}$$

- Formula 2 (when you have potential energy)

$$V_f - V_i = \frac{U_f - U_i}{q}$$

- Formula 3 (going from *i* to *f* in the presence of **E**)

$$V_f - V_i = -\int_i^f \mathbf{E} \cdot d\mathbf{r}$$

- Formula 4 (when dividing a continuous extended object into small point charges and then integrate)

$$V = \frac{1}{4\pi\epsilon_0}\int \frac{dq}{r}$$

Electric Potential

- Total potential is always the sum of individual potentials: either integral for continuous and sum for discrete

$$V = V_1 + V_2 + \cdots$$

Gradient

- Gradient in Cartesian coordinates

$$\nabla = \hat{\mathbf{i}}\frac{\partial}{\partial x} + \hat{\mathbf{j}}\frac{\partial}{\partial y} + \hat{\mathbf{k}}\frac{\partial}{\partial z} \Rightarrow \nabla f = \hat{\mathbf{i}}\frac{\partial f}{\partial x} + \hat{\mathbf{j}}\frac{\partial f}{\partial y} + \hat{\mathbf{k}}\frac{\partial f}{\partial z}$$

- Gradient in spherical coordinates

$$\nabla = \hat{\mathbf{r}}\frac{\partial}{\partial r} + \hat{\boldsymbol{\theta}}\frac{1}{r}\frac{\partial}{\partial \theta} + \hat{\boldsymbol{\varphi}}\frac{1}{r\sin(\theta)}\frac{\partial}{\partial \varphi} \Rightarrow \nabla f = \hat{\mathbf{r}}\frac{\partial f}{\partial r} + \hat{\boldsymbol{\theta}}\frac{1}{r}\frac{\partial f}{\partial \theta} + \hat{\boldsymbol{\varphi}}\frac{1}{r\sin(\theta)}\frac{\partial f}{\partial \varphi}$$

- In any coordinates

$$df = \nabla f \cdot d\mathbf{r}$$

- So in Cartesian

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz$$

Gradient of Potential

- We have

$$\mathbf{E} = -\nabla V \Rightarrow \begin{cases} E_x = -\frac{\partial V}{\partial x} \\ E_y = -\frac{\partial V}{\partial y} \\ E_z = -\frac{\partial V}{\partial z} \end{cases}$$

- Based on previous slide, conversely we have

$$dV = -\mathbf{E} \cdot d\mathbf{r}$$

- In any coordinates

$$df = \nabla f \cdot d\mathbf{r}$$

- So in Cartesian

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

Capacitors

- Capacitance

$$C = \frac{Q}{V}$$

- For parallel plate capacitor we also have

$$C = \epsilon_0 \frac{A}{d}$$

- Electric field and potential

$$V = Ed$$

Capacitors

- In series

$$
\begin{aligned}
Q &= Q_1 = Q_2 \\
V &= V_1 + V_2 \\
\frac{1}{C} &= \frac{1}{C_1} + \frac{1}{C_2} \Rightarrow C = \frac{C_1 C_2}{C_1 + C_2}
\end{aligned}
$$

- In parallel

$$
\begin{aligned}
Q &= Q_1 + Q_2 \\
V &= V_1 = V_2 \\
C &= C_1 + C_2
\end{aligned}
$$

Energy in Capacitors

- Potential energy stored in a capacitor

$$U = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}QV = \frac{1}{2}CV^2$$

- Energy density (for any electric field and for capacitors

$$u = \frac{dU}{d\mathcal{V}}$$
$$u = \frac{1}{2}\epsilon_0 E^2$$

Dielectrics

- Dielectric constant

$$K = \frac{C}{C_0}$$

- Dielectric constant if the charge Q on conductor plates remains the same after inserting the dielectric

$$K = \frac{V_0}{V}$$

- Electric field

$$|\mathbf{E}| = \frac{|\mathbf{E}_0|}{K}$$

Dielectrics

- Polarized induced charge on the sides of dielectric in a capacitor

$$\sigma_i = \sigma_0 \left(1 - \frac{1}{K}\right)$$

where σ_0 is the original charge (that does not change) and the total charge on one side on a conductor plate and on the dielectric is

$$\begin{cases} \text{on the positive side: } \sigma_0 - \sigma_i \\ \text{on the negative side: } -\sigma_0 + \sigma_i \end{cases}$$

Dielectrics

- Permittivity of dielectric

$$\epsilon = K\epsilon_0$$

- Field inside a parallel plate capacitor with a dielectric

$$|\mathbf{E}| = \frac{\sigma_0}{\epsilon}$$

- Capacitance

$$C = KC_0 = \epsilon\frac{A}{d}$$

- Energy density

$$u = \frac{1}{2}\epsilon E^2$$

Dielectrics

- Gauss's law in with dielectrics

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{in-free}}}{\epsilon}$$

where $q_{\text{in-free}}$ is the free charge, i.e., only the charge on the conductor, not the polarized induced charge on dielectric

$$|\mathbf{E}| = \frac{\sigma_0}{\epsilon}$$

Sample problem 1

A solid metal sphere with radius r_a is located at the center of a hollow, metal, spherical shell with radius r_b. Note that the two objects share the same center, and $r_a < r_b$. There is a total charge of $+q$ on the inner sphere and a total charge of $-q$ on the outer spherical shell. (a) Calculate the potential $V(r)$ for (i) $r < r_a$; (ii) $r_a < r < r_b$; (iii) $r > r_b$. Take V to be zero when r is infinite.

Sample problem 1

Sample problem 1

Sample problem 1

Sample problem 1

Sample problem 1

Sample problem 1

Sample problem 1

Sample problem 1

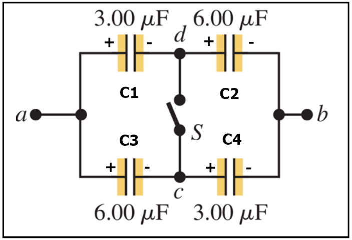
Sample problem 1

Sample problem 1

Sample problem 1

Sample problem 2

The capacitors in the figure below are initially uncharged and are connected, as shown in the diagram, with switch S open. The applied potential difference across the terminals a and b is
$V_{ba} = V_a - V_b = 210$ V.
(a) What is the potential difference $V_{dc} = V_c - V_d$ when switch S is open.
(b) What is the potential difference across each capacitor after switch S is closed?
(c) How much charge flowed through the switch when it was closed?



Sample problem 2

Sample problem 2

Sample problem 2

Sample problem 2

Sample problem 2

Sample problem 2

Sample problem 2

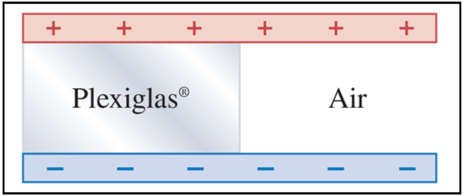
Sample problem 3

A parallel-plate capacitor is made from two square plates 12.0 cm on each side and 4.50 mm apart. Half of the space between these plates contains only air, but the other half is filled with Plexiglas of dielectric constant 3.20 (see figure below). A 23.0 V battery is connected across the plates.

(a) What is the capacitance of this combination? (Hint: Can you analyze the system as several capacitors in parallel and/or series?)

(b) How much energy is stored in the capacitor?

(c) If we remove the Plexiglas but change nothing else, how much energy will be stored in the capacitor?



Sample problem 3

Sample problem 3

Sample problem 3